

Assignment 3

1 Neural Machine Translation with RNNs (45 points)

- (g) *enc_mask* marks padding tokens with 1s and other non-padding tokens with 0s. And the attention scores of tokens where *enc_mask* is 1 are set to $-\infty$. Therefore, with *enc_mask*, model does not attend to padding tokens since their attention values are 0 after softmax. Such masks are necessary for NMT models to attend to meaningful tokens of a sentence, rather than padding tokens that are manually added to support parallel processing.
- (h) Corpus BLEU: 19.915406403991692.
- (i) Pros of dot production attention: computationally effective.
Cons of dot production attention: assumes the two vectors are in the same space (aligned).
Pros of additive attention: adds non-linearity and is more expressive.
Cons of additive attention: more computational overhead.

2 Analyzing NMT Systems (25 points)

- (a) The convolution layer after embedding combines individual token embeddings, which gives model better semantic representations of phrases consisting of multiple tokens.
- (b)
 - i. The NMT model translates the phrase 贼人 into the singular form *the culprit was* while the gold reference is the plural form *the culprits were*. The error occurs due to morphological ambiguity in the source language (Chinese), where the noun 贼人 does not explicitly indicate grammatical number (singular vs. plural). The model likely defaults to the singular form based on its frequency in the training data, as it lacks sufficient contextual clues or explicit number markers to infer the plural sense. To fix such errors, nouns in the source corpus could be pre-processed with explicit entity number tags.
 - ii. The model repeats *resources have been exhausted* and omits *no space to accommodate these people*. This error may originate from attention mechanisms. The model allocate too much attention on the second part 资源已经用尽. A possible fix is to change the attention mechanism during decoding.
 - iii. The model translates 今天是国殇日 into *today's day*, seeming to take the source as a clause, and 国殇 is omitted in translation. In training data, 国殇 appears only once, thus it's hard for the model to learn a robust representation for this word. To fix, more sentences including 国殇 should be added into the corpus.
 - iv. The model ignores the meaning of 唔做 during translation. It seems that the model doesn't understand the causal relationship between 唔做 and 唔错. The main reason for this may be lack of Cantonese showing such causal relationship in the training

corpus. A possible fix for this can be adding more Cantonese texts with more complicated logics into the training data.

- (c) i. $\mathbf{c}_1$: $p_1 = \frac{4}{9}, p_2 = \frac{3}{8}, \text{len}(c_1) = 9, \text{len}(r = r_1) = 11, BP = \exp\left(-\frac{2}{9}\right), BLEU \approx 0.327$.
 $\mathbf{c}_2$: $p_1 = 1, p_2 = \frac{3}{5}, \text{len}(c_2) = 6, \text{len}(r = r_2) = 6, BP = 1, BLEU \approx 0.775$.
 $\mathbf{c}_2$ is considered to be a better translation according to the BLEU score. But it's obvious that $\mathbf{c}_1$ is better indeed.
- ii. $\mathbf{c}_1$: $p_1 = \frac{4}{9}, p_2 = \frac{3}{8}, \text{len}(c_1) = 9, \text{len}(r = r_2) = 6, BP = 1, BLEU \approx 0.408$.
 $\mathbf{c}_2$: $p_1 = \frac{1}{2}, p_2 = \frac{1}{5}, \text{len}(c_2) = 6, \text{len}(r = r_2) = 6, BP = 1, BLEU \approx 0.316$.
 $\mathbf{c}_1$ is considered to be a better translation according to the BLEU score.
- iii. As we can see above, different numbers of reference sentences lead to different BLEU results. BLEU just matches n-grams between the reference sentence(s) and the translation. When there are multiple sentences, the model can achieve a high score by generating words that are scattered across different references, thus forming a grammatically incorrect sentence that hacks the evaluation metric.
- iv. Advantages: fast and computationally efficient; objective and reproducible.
 Disadvantages: blind to syntax, grammar, and semantics; correlates poorly with human judgment.
- (d) i,ii. The translation quality improve over the training iterations. See coding/outputs for examples.